AC9M6M02 • Year 6 Mathematics

Establishing and Applying the Rectangle Area Formula

Connect arrays, multiplication, units and composite practical regions

We are learning to…

Students derive area from rows and columns of square units, explain A = length × width and apply it to missing dimensions, composite spaces and material estimates.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

establish the formula for the area of a rectangle and use it to solve practical problems

Derive $A = l \times w$ from a square-unit array

rectangle	rows	squares per row	area
6 cm × 4 cm	**4**	**6**	**24 cm²**
$l \times w$	w rows	l per row	lw square units

The formula summarises repeated counting of square units. Units are squared because area measures two-dimensional coverage.

Solve practical and missing-dimension problems

floor 8 m × 5.5 m
44 m²

area 72 m², width 8 m
length = 9 m

L-shape
large rectangle − missing rectangle

tiles
area ÷ tile area; round for whole tiles and waste

Check dimensions use compatible units and distinguish area, perimeter and material quantity.

Build and annotate the model

Represent establishing and applying the rectangle area formula and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Perimeter formula used:** Area counts covering; perimeter traces boundary.
- **Units not squared:** Use cm^2, m^2 and related square units.
- **Length and width in different units:** Convert before multiplying.
- **Tile count left fractional:** Interpret whole items and allowance.

Quick check

- Derive 6×4 area.
- Find a missing dimension.
- Decompose an L-shape.
- Convert before area.
- Interpret tile count.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.